Temporary Address
State College, PA 16801

Evan Chilek

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Permanent Address
Scranton, PA 18403

Objective

Junior mechanical engineering student in the Schreyer Honors College looking for an internship for Summer 2027. Experience in CAD modeling, computational analysis, and manufacturing through projects and as a mechanical intern designing building systems and creating technical documentation.

Education

The Pennsylvania State University, University Park, PA — GPA: 3.97
B.S. in Mechanical Engineering with honors — Expected: May 2028

Employment History

Mechanical Intern, Greenman-Pedersen Inc., Scranton, PA — 5/25-8/25; 5/26-8/26

- Performed heating and cooling load calculations using Trace 700 to size HVAC equipment and develop mechanical systems, ensuring compliance with performance and ASHRAE standards
- Developed detailed mechanical drawings, BIM models, and created schedules in Revit and AutoCAD for commercial and institutional building projects

Cashier, Wegmans, Dickson City, PA — 11/20-Present

Projects

Mu-Bot Research, BioRob InFL Lab — Fall 2026

- Undergraduate research assistant on a project with the goal of developing autonomous robotic swimmer
- Exploring and testing how different materials and mechanical designs affect the elasticity and modularity of the body to maximize performance in bursts and efficiency in coasting.

Small-Scale Motorjet Propulsion System, Personal Project — Summer 2026

- Designed and fabricated a small-scale motorjet propulsion system, integrating compressor, combustion chamber, fuel delivery, ignition, and nozzle subsystems into a functional test platform
- Conducted experimental testing and performance evaluation to assess combustion stability, airflow characteristics, thermal behavior, and thrust generation while iterating designs to improve performance

Enterprise Drone, Manufacturing Lead, ASME Design Team — Fall 2025- Spring 2026

- Performed testing and conducted finite element analysis on Fusion 360 to drive iterative design improvements to increase performance by reducing weight and maintain a factor of safety
- Selected materials such as CFRP, PLA, and aluminum by performing trade-off analysis considering mechanical properties, cost, and manufacturability to support design decisions

Technical Skills

Certification: Autodesk Certified Associate in CAD for Mechanical Design — ADSK-159-1196
Modeling/Analysis: Fusion 360, SolidWorks, AutoCAD, Revit/MATLAB, Fusion 360(simulation)

Honors

Wabtec Design Competition: Most Sustainable Design